Leo Marcinkus

ECEN 5224

Week 2 Lab Report

01/21/26

Lab 1

Objective:

The objective of this lab is learn ADS and understand component behavior by creating a circuit to calculate the impedance of a 2-inch transmission line. By varying its port between an open and short an inductor and capacitor's measured impedance can be matched to the transmission line. An ideal T-model will also be simulated. Results should be compared to what is expected and the final outcome should be analyzed.

Method:

ADS was used to simulate the created circuits. A two-port parameter S-block was used with a 2-inch transmission line .s2p file imported into it. Port 1 had an AC current source flowing into it with port 2 connected to a resistor. The resistor being of value $1e^{9}$ Ω if it is open or $1e^{-6}$ Ω if it is shorted. The AC sweep was run from 20 MHz up to 10 GHz.

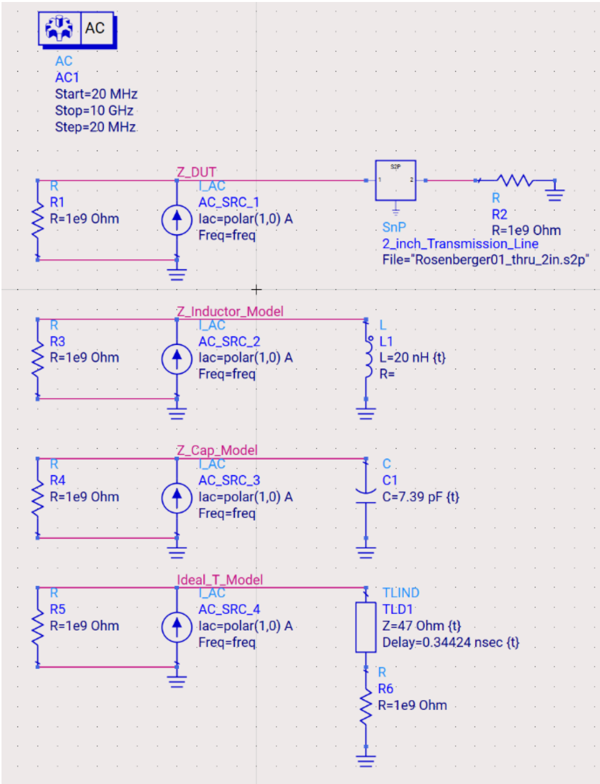

Figure 1: Lab 1 Completed Circuit

L-element Impedance Matching:

The transmission line is shorted and the inductor's value is tuned to match the impedance. I know that an L element's impedance is modeled by $Z_L = j\omega L$ meaning a larger inductance results in a larger impedance. This helped me match the impedance by setting the inductor to a value of 20 nH.

I would expect the inductor to match well at lower frequencies. As the frequency starts to climb, I expect a much larger difference. Using a general rule of thumb and based on what was demoed in class, I would expect a decent match up to around 200 Mhz.

C-element Impedance Matching:

The transmission line is opened and the capacitor's value is tuned to match the impedance. I know that a C element's impedance is modeled by $Z_C = \frac{1}{j\omega C}$ meaning a larger capacitance results in a smaller impedance. The tuned capacitor value came out to be 7.39 pF. This resulted in a well-matched impedance compared to the open transmission line.

Similar to the L-element, I expect the capacitor to match the behavior of the transmission line at lower frequencies and then start to deviate. I expect a decent match of this impedance up to around 200 MHz just like the inductor.

Ideal T-Model Impedance Matching:

The circuit is opened for both the transmission line and the ideal T-block. Similar to before, the T-element is tuned to match the impedance of the transmission line. Great values came out to be Z = 47 Ω with a delay of 0.34424 nsec. This lines up with expectations knowing that a common Z value tends to be 50 Ω.

Results and Plots:

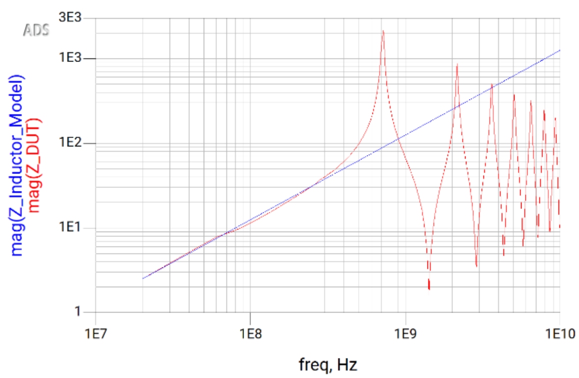

Figure 2: L-Element 20nH Matched Impedance

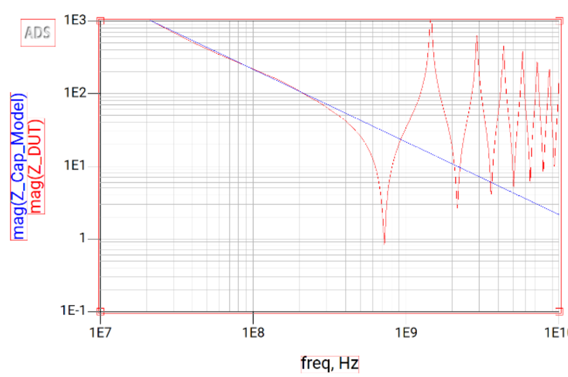

Figure 3: C-Element 7.39pF Matched Impedance

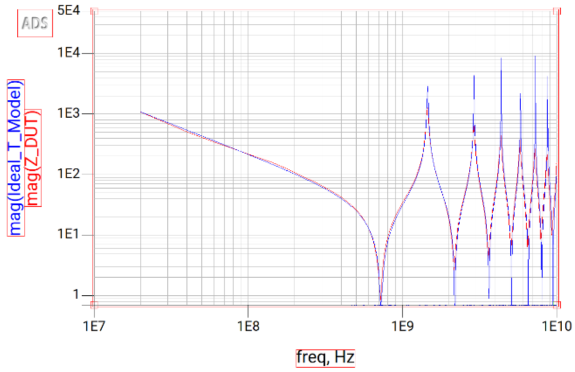

Figure 4: Ideal T-Element 47Ω with 0.34424 nsec delay Matched Impedance

Analysis and Conclusion:

My expectations for the behavior of each model are largely supported. However, looking closer at the plots it is clear that the behavior for the inductor and capacitor start to deviate slightly earlier than I expected. Both elements match well up to the around 150 MHz which is a little lower than my anticipated 200 MHz estimation. The ideal T-model shows almost perfect impedance matching all the way to 10 GHz. Analyzing the plot further, it is seen that the maximum and minimums deviate more from the transmission line as the frequency continues to increase.

These results show that simple lumped models, like these inductor and capacitor examples, can provide a great approximation of impedance of a transmission line in lower frequencies. However, this is only up to the first couple hundred MHz as measured above. Unlike the L and C models, the ideal T-element model can sustain a far more accurate match at much higher frequencies. It is clear that the model being used for a specific purpose should have its size, length, and optimal frequency considered. For a shorter interconnect, a simple lumped element model will suffice. But if higher frequencies are utilized, an ideal distributed transmission line model will prove more worth.

Lab 2

Objective:

The objective of this lab is to understand component behavior by creating a circuit to calculate the impedance of an EM simulated via. By varying its port between an open and short an inductor and capacitor's measured impedance can be matched to the simulated via. An ideal T-model will also be simulated. Results should be compared to what is expected and the final outcome should be analyzed.

Method:

ADS was used to simulate the created circuits. A two-port parameter S-block was used with a simulated via .s2p file imported into it. Port 1 had an AC current source flowing into it with port 2 connected to a resistor. The resistor being of value $1e^{9}$ Ω if it is open or $1e^{-6}$ Ω if it is shorted. The AC sweep was run from 20 MHz up to 10 GHz.

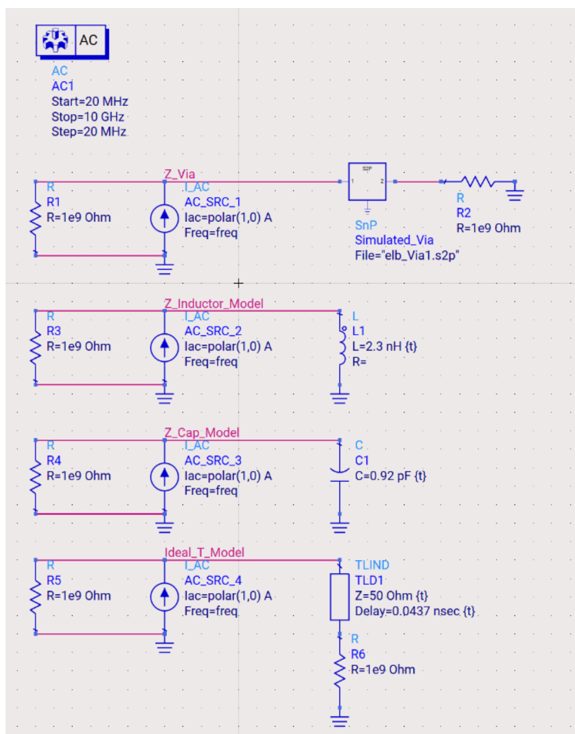

Figure 5: Lab 2 Completed Circuit

L-element Impedance Matching:

The simulated via is shorted and the inductor's value is tuned to match the impedance. I know that an L element's impedance is modeled by $Z_L = j\omega L$ meaning a larger inductance results in a larger impedance. This helped me match the impedance by setting the inductor to a value of 2.3 nH.

I would expect the inductor to match well at lower frequencies, but higher than the transmission line. As the frequency starts to climb, I expect a deviation. Since the length of the via is even shorter than the transmission line, I would expect this inductor to remain valid at even higher frequencies than before of 200-300 MHz.

C-element Impedance Matching:

The simulated via is opened and the capacitor's value is tuned to match the impedance. I know that a C element's impedance is modeled by $Z_C = \frac{1}{j\omega C}$ meaning a larger capacitance results in a smaller impedance. The tuned capacitor value came out to be 0.92 pF. This resulted in a well-matched impedance compared to the open simulated via.

Similar to the L-element, I expect the capacitor to match the behavior of the simulated via at frequencies of 200-300 MHz.

Ideal T-Model Impedance Matching:

The circuit is opened for both the simulated via and the ideal T-block. Similar to before, the T-element is tuned to match the impedance of the transmission line. Great values came out to be Z = 50 Ω with a delay of 0.0437 nsec.

Results and Plots:

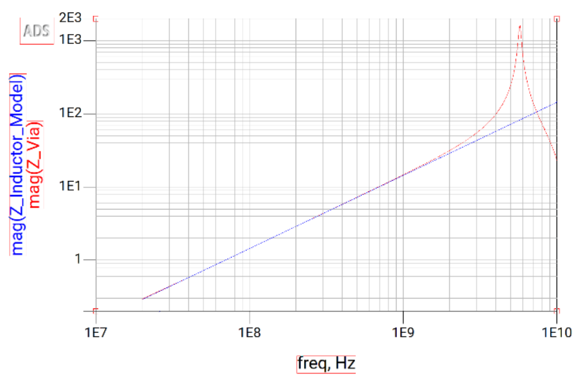

Figure 6: L-Element 2.3nH Matched Impedance

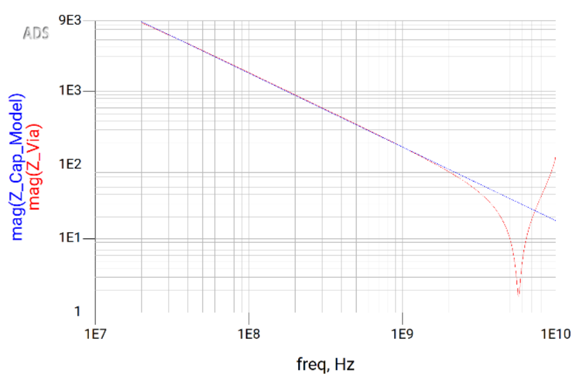

Figure 7: C-Element 0.92pF Matched Impedance

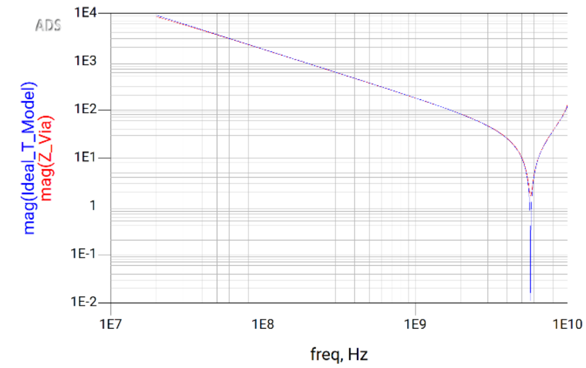

Figure 8: Ideal T-Element 50Ω with 0.0437 nsec delay Matched Impedance

Analysis and Conclusion:

My expectations for the behavior of each model are largely supported. Looking closer at the plots, the behavior for the inductor and capacitor start to deviate right where I anticipated at around the 200 MHz estimation. The ideal T-model shows almost perfect impedance matching all the way to 10 GHz. Analyzing the plot, it is seen that the minimum deviates more from the simulated via in the T-model but nothing further.

These results show that simple lumped models, like these inductor and capacitor examples, can provide a better approximation for a simulated via compared to a transmission line. This is due to their ability to remain accurate at higher frequencies for shorter interconnects. This is unlike the previous behavior when matched to a 2-inch transmission line. Similar to before, the ideal T-element model can sustain a far more accurate match at even higher frequencies than the lumped models. These results further emphasize the importance on selecting what model is most optimal for a specific task. For a simulated via, a simple lumped element model will remain effective for higher frequencies than before. But if the limit is exceed, an ideal distributed transmission line model will still prove more worth.